

DRAFT, sections 1-7 (for review; not for deposit)

A Blind Benchmark of Decipherment Methods on Synthetic Scripts Calibrated to Indus Statistics

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This work does not decipher, and does not claim to decipher, the Indus script.

Scope: in this paper "decipherment" means substitution-style recovery of sign values only.

Conventions used throughout. Unless a sentence says otherwise, a recovery number is Task D token accuracy under the **original (frozen) EM selection rule**, scored **before the cognate step**. Scores that include the solver's cognate step are secondary and are labelled "with oracle sound correspondences". The revised selection rule was adopted after the original rule had been run and is reported as post hoc. Intervals are 95% and resample whole source languages.

Abstract

Many arguments about the undeciphered Indus script rest on statistics or decipherment methods that have not been tested on corpora of Indus size and shape with a known answer. We build synthetic corpora from five known languages and from non-linguistic sign systems, write them in invented scripts, calibrate them to published Indus statistics (2,906 texts, 4.6 signs per text), and hide the answer keys. We test eight method families, frozen before the final runs, under three levels of solver knowledge. Four results replicate on fresh seeds. (1) Entropy statistics do not separate synthetic languages from independent random signs at Indus scale. (2) Without a related language among the candidates, mean recovery of sign values stays at or below 4.8% in every run. (3) At equal total size, longer inscriptions help more than more inscriptions, in both generators and in logo-syllabic corpora. (4) Published Indus statistics do not fix how decipherable an Indus-scale corpus is: two generators that both match them differ roughly three- to fivefold when a related language is available, and for logo-syllabic corpora this contrast appears only when the related language is the sole reference. A fifth, archaeological result is a prediction, not a measurement. The main solver matches the best published accuracy on English letter-substitution ciphers. Limitations, including an optimistic synthetic relative, few source languages and post-hoc decisions, are listed in full. This work does not decipher, and does not claim to decipher, the Indus script.

Plain-language summary

The Indus script, used in South Asia about 4,500 years ago, has never been read. Its inscriptions are very short, and no text repeats an Indus message in a readable script. We asked: could today's computer methods read writing like this if the answer were known? We wrote five known languages in invented scripts, matched them to published Indus statistics, and hid the key. The main method, expectation-maximisation (EM), guesses what each sign stands for, checks how well the guesses fit a known language, and improves them. We score token accuracy: the share of all signs in the texts that a method reads correctly. Methods were given either a close relative of the hidden language (related), a relative mixed with unrelated languages (candidates), or unrelated languages only (none). Without a relative, methods read about 5% of signs or fewer on average.

With one, results depended strongly on corpus properties that published statistics leave open.

1. Introduction

The Indus inscriptions are short. The concordance of Mahadevan (1977) holds 2,906 texts with 13,372 sign occurrences, about 4.6 signs per text (Yadav et al. 2010; Rao 2018), and the longest text is commonly given as 17 signs on a single surface (Farmer, Sproat & Witzel 2004). Published sign lists count between 386 and about 700 sign types, depending on how variants are grouped (Parpola 1994 and Wells 2015, both via Rao 2018; Mahadevan 1977; Wells 2006 via Yadav et al. 2010). No bilingual text is known, and no language of the Harappan civilisation is agreed.

Two kinds of argument have been made about this corpus. Some assign readings to the signs. Others use corpus statistics to argue that the signs do, or do not, encode language: conditional entropy (Rao et al. 2009), n-gram models (Yadav et al. 2010), and, on the other side, text brevity (Farmer, Sproat & Witzel 2004) and the finding that entropy-type measures do not separate writing from non-linguistic symbol systems (Sproat 2010, 2014). Both kinds of argument depend on a further question: would the method work on a corpus of Indus size and shape whose answer is known?

We test that question directly. We write known languages, and sign systems that encode no language, in invented scripts. We calibrate each synthetic corpus to published Indus statistics, hide its answer key, and measure what existing methods recover. The goal is a statement of the form "on corpora that match the published Indus statistics, method M recovers X% of sign tokens under conditions K". Such a statement says what a method can do on Indus-like data. It says nothing about what the Indus signs mean.

Two scope statements apply throughout. First, "decipherment" here means substitution-style recovery of sign values only: assigning each sign a sound, syllable or word of a known reference language. It does not cover grammar, meaning, or identifying an unknown language. Second, the benchmark languages (Sanskrit, Old Tamil, Sumerian, Latin, Finnish) were chosen for open or documented licences and for spread across language families and script types. Their use implies nothing about the language or languages of the Indus inscriptions, and how well any of them is recovered says nothing about Harappan.

Defining "Indus-like" turned out to be the central difficulty. Published statistics fix the number of texts, their length, the sign inventory and several frequency and positional measures. They do not fix how often texts repeat, how many sign types are variants of one another, or how text material is sampled. We find that choices these statistics leave open, here the way text is sampled, change recovery roughly three- to fivefold when a related language is available, while duplicate rate and inventory within one generator matter less under the frozen rule. The paper therefore reports a map over them rather than a single Indus number.

Findings. (1) Entropy statistics do not separate synthetic languages from independent, identically distributed (i.i.d.) signs at Indus scale. (2) Without a related language among the candidates, mean recovery stays at or below 4.8% in every run; no corpus reaches 50% of tokens; the best single corpus reaches 28%. (3) Under the frozen rule, longer inscriptions help more than more inscriptions in both generators, including logo-syllabic

corpora (12.7% vs 0.2% at equal total tokens, generator v1). (4) Published Indus statistics do not fix how decipherable an Indus-scale corpus is. Two generators that both match them differ roughly three- to fivefold when a related language is available. For logo-syllabic corpora, the most Indus-relevant type, this contrast appears only when the related language is the sole reference; among candidate languages it is unresolved. (Frozen rule, no oracle correspondences: 2.4% vs 12.9% of sign tokens on the main seeds, 5.3x; 2.7% vs 9.4% on fresh seeds, 3.5x; 2.6% vs 9.3% with the stricter sister-v3, 3.7x.) (5) Archaeologically defined subsets of the corpus are predicted to sit at different points on the map; these are predictions, not measurements.

What this paper provides. (i) A generator of synthetic scripts calibrated to published Indus statistics, with every target cited and flagged as verified or not. (ii) Eight method families behind one interface, frozen before the final runs. (iii) A protocol that scores four tasks separately, with intervals that resample whole source languages, three calibration regimes, three levels of solver knowledge and replication on fresh seeds. (iv) A sensitivity map over duplicate-text rate and sign inventory, with published reference points marked. (v) A blind challenge with hash-committed answer keys.

2. Related work

Structural statistics of the Indus corpus. Rao et al. (2009) compared the conditional entropy of Indus sign sequences with linguistic and non-linguistic reference systems and argued for linguistic structure. Yadav et al. (2010) fitted n-gram models to the corpus, reported its frequency and positional statistics, and are the source of most of our calibration targets. Farmer, Sproat & Witzel (2004) argued from text brevity and sign repetition that the signs were not a writing system. Sproat (2010, 2014) showed that entropy-type statistics do not reliably separate writing from non-linguistic symbol systems; Lee, Jonathan & Ziman (2010) proposed an entropy-based classifier for the Pictish symbols that we include as a method. Our Task A asks whether these statistics work at Indus scale when the answer is known.

Closest prior work. Nair (2026, arXiv:2604.17828, a preprint) tests the observed Indus corpus (1,916 deduplicated ICIT inscriptions) against computer-generated non-linguistic baselines (a heraldic and an administrative generator) and seven attested non-linguistic corpora, on four properties from the Farmer-Sproat-Witzel critique, and places the corpus between the baselines. Tiwari (2026, NLP4DH, pp. 314-319) analyses 6,579 inscriptions with visual clustering of sign forms, entropy, Kullback-Leibler divergence and a BiLSTM, and reports directional asymmetry and structured combinatorial patterns. Both papers study the real corpus and ask what kind of sign system it is. This paper does not analyse the real corpus. It generates synthetic corpora with known answers at Indus scale, linguistic and non-linguistic, and measures what existing methods recover from them; its question is about the methods, not the Indus signs. Nair's synthetic baselines are compared with the real corpus; ours are scored against their own hidden keys. We use Nair's ICIT duplicate rate only as a reference point.

Computational decipherment. Knight et al. (2006) framed decipherment as unsupervised learning of a substitution model with expectation-maximisation (EM); Berg-Kirkpatrick & Klein (2013) showed that many random restarts improve such solvers. Snyder, Barzilay & Knight (2010) recovered Ugaritic using Hebrew as a known relative, and Luo, Cao & Barzilay (2019) recovered Ugaritic and Linear B cognates with a neural

model and minimum-cost flow. Each success had a closely related known language and far more text per document than the Indus corpus. Our knowledge tiers separate these conditions. The EM cognate-matcher we test borrows only the one-to-one assignment idea of Luo, Cao & Barzilay (2019); it contains no neural network, and its results say nothing about neural methods.

Script-type estimation and segmentation. We estimate script type from inventory size, with a Chao (1984) richness estimate, and segment with branching entropy (Harris 1955; Tanaka-Ishii 2005).

Archaeological context. Kenoyer & Meadow (2010) and Kenoyer (2020a, 2020b) describe how inscribed objects at Harappa differ by type and period: seals are almost all unique, while incised and molded tablets often occur as copies or same-mold duplicates, and the script was used and changed over roughly 700 years. We use these statements only qualitatively, to predict where subsets of the corpus would sit on our map (Section 4.5).

3. Benchmark design

3.1 Sources and licences

Plaintext comes from five languages in four families: Sanskrit (Rāmāyaṇa, Digital Corpus of Sanskrit, CC BY 4.0), Old Tamil (17 Sangam root texts, Project Madurai), Sumerian (CDLI bulk ATF dump, including Ur III seal inscriptions), Latin (Caesar and Vergil, Project Gutenberg) and Finnish (Kalevala, Project Gutenberg). CDLI's terms allow reuse "according to common and fair academic practice" with citation, which is not an open licence; Sumerian-derived material is therefore used in internal experiments only and excluded from published challenge rounds. No source text is redistributed. Each source is cut into short, seal-like clauses (names, epithets, titles, formulae and, for Sumerian, real seal inscriptions).

3.2 Synthetic scripts

Words are written in one of four script types: alphabetic (phonemes), syllabic (CV, V and C units), logographic (word forms) and logo-syllabic (the 250 most frequent words as logograms, other words spelled syllabically). Units map to random sign identifiers. Optional phenomena are allographs, homophones, polyvalent signs, determinatives, word dividers and reading direction.

3.3 Calibration targets

Every target is listed in `config/indus_targets.yaml` with its citation, tolerance, the size of the corpus it was measured on, and whether we read the number in the source ourselves. The Indus point is 2,906 texts with mean length 4.60 (Mahadevan 1977, via Yadav et al. 2010 and Rao 2018). Other targets are the longest text (commonly given as 17 signs on a single surface; Farmer, Sproat & Witzel 2004), the sign inventory (400-700), the number of signs covering 80% of tokens (69), the share of the most frequent sign (0.10), and the numbers of text-final and text-initial signs covering 80% of those positions (23 and 82), all from Yadav et al. (2010). The median length (4) is derived from Yadav et al. 2010, Fig. 2 (2,591 texts, mean 3.92, max 14; not like-for-like with 2,906 texts at 4.6 signs). It was originally unsourced (Appendix: analysis history). The generator also uses it to set the spread of text lengths. The hapax share of sign types (0.27, Possehl 2002) is read only via

a secondary source. Size-dependent statistics are computed on subsamples of the reference corpus size.

The duplicate-text rate is not a target, because no source fixes it for the corpus as a whole. We report two published values as reference points: 0.354 for Mahadevan's concordance, read from the vector data of Yadav et al. (2010, Fig. 2; 0.281 without its four most repeated texts), and 0.237 for the ICIT corpus, from a preprint we could not verify (Nair 2026). The figure in Yadav et al. plots 2,591 texts, not 2,906; the paper does not say which texts are left out.

3.4 Calibration regimes

Three regimes control how much of the target set is imposed. `full` imposes every target. `holdout` imposes only size, length and inventory, leaving the frequency and positional statistics that Task A methods measure free, so that a method cannot be rewarded for statistics we injected. `wrong_prior` imposes deliberately wrong targets (longer texts, smaller inventory). Results lead with `holdout` wherever a method could exploit calibration.

3.5 Non-linguistic controls

The controls encode no language: heraldic bearings generated with the rule of tincture (one sign per visual element), administrative slot tags, sparse Markov emblems with positional preferences, the rigid and random-order controls of Rao et al. (2009) (they call the random-order system type 1 and the rigid one type 2; our internal keys `rao_type1` and `rao_type2` use the opposite numbering, so this paper names them by behaviour), and an adversarial Markov chain tuned to match the entropy ratio of the language corpora. Japanese *kamon* descriptions are text and are reported separately as a contaminated control.

3.6 Knowledge tiers and the synthetic sister language

Tasks C (language family) and D (sign values) are run at three levels of solver knowledge. `related` gives the solver one reference language, a synthetic sister of the hidden language. `candidates` gives the sister together with every other benchmark language, so the solver must choose. `none` removes the hidden language and its sister. Only `candidates` and `none` are relevant to the Indus case, because no close relative of the Harappan language is agreed, and we do not know which of the two applies.

The sister is built from the half of the source text not used to generate the hidden corpus. A seeded 30% of consonants and of vowels are permuted by a regular, bijective correspondence, and 20% of word types are replaced by unrelated words of the same length. **The sister is therefore much closer to the hidden language than any attested relative of an undeciphered script is likely to be, in three measurable ways:**

1. **Oracle sound correspondences.** When the solver selects the sister, its frozen cognate step maps sister units to hidden-language units with the true correspondence table. We report every sister-tier score before this step. With the step, scores rise by 7-8 points (generator v1) and 17-19 points (generator v2) in the related tier at the Indus point, and by up to 51 points on a single corpus. Those higher scores appear only as secondary numbers, labelled "with oracle sound correspondences".

2. **Shared text material: residual verbatim overlap.** The sister comes from the same source text. In the first version of the benchmark (sister-v1) the clauses were split into the two halves by position, so a clause that occurs more than once could fall in both halves. We corrected this (sister-v2): clauses are now split by a hash of their content, and no clause occurs in both halves. The correction moved no Indus-point mean outside its sister-v1 interval. **But word-for-word overlap remains high.** At the Indus point, the share of hidden texts that occur verbatim inside some clause of the sister's half averages 30-53% by language under generator v1 (Sumerian 30%, Tamil 36%, Latin 36%, Sanskrit 46%, Finnish 53%) and 43-56% under generator v2 (Latin 43%, Tamil 46%, Finnish 48%, Sumerian 55%, Sanskrit 56%), over 12 corpora per language (4 script types, 3 seeds); single corpora range from 8% to 90%. Short phrases recur throughout a text in one language, so splitting by clause cannot remove them. A real relative would share far fewer exact phrases, since vocabulary, morphology and word order diverge. All sister-tier results in this paper carry this caveat.
3. **No structural divergence.** The sister keeps the hidden language's word order and morphology; only phonemes and a fifth of the vocabulary change. The sister's distance is a free parameter; a sweep over four distances is reported in the appendix.

Section 4.6 repeats the key comparisons with a stricter sister (sister-v3), which drops every sister clause that contains any hidden text's word sequence; this removes 33-42% of the sister's clauses at the Indus point and leaves 7-8% of the hidden texts' word pairs in it (31-35% under sister-v2). Sister-v1 results are reported only as "upper bound (sister-v1, known overlap)".

3.7 Methods, frozen

Eight method families run on every corpus behind one interface: conditional and block entropy (after Rao et al. 2009), a bigram Markov model (after Yadav et al. 2010), positional histograms, segmentation, script-type estimation by inventory rule and by classifier, a two-parameter tree (Lee, Jonathan & Ziman 2010), a multi-feature language classifier, Knight-style EM, an EM cognate-matcher and a frequency-rank baseline. All method code was frozen at the git tag `frozen-v1` before the final runs; later changes touch only data generation and analysis.

EM needs a rule for choosing among candidate reference languages. The rule written before the runs (the **original rule**) selects by raw likelihood with learned emissions for a pooled rare-unit state. After the original rule selected Sumerian for almost every corpus, we wrote a **revised rule** (uniform emissions for the pooled state; selection by bigram gain normalised by the reference's own mutual information). The original rule is primary throughout; the revised rule is reported as post hoc.

3.8 Tasks and scoring

Task A asks whether a corpus encodes language (balanced accuracy; chance 0.5). Task B asks for the script type (accuracy; chance 0.25). Task C asks for the language family (accuracy against chance of one over the number of candidate families). Task D asks for the value of each sign and is scored as the share of sign tokens whose predicted value equals the true value; "success" is at least 50%. Decision rules for Tasks A and B are learned leave-one-source-out, so a Sanskrit corpus is judged by a rule fitted only on other

sources. Means carry 95% cluster-bootstrap intervals that resample source languages (or control families) and then corpora within them, because corpora from one source are not independent. A shuffled-prediction control checks that the Task D scorer is not generous: permuting a solver's predicted values across signs drops a 93% corpus to 0.9%. Positive controls against published results are reported in Section 4.9.

Corpora are generated at three seeds per configuration (0-2) and replicated on fresh seeds (3-5). Two generators are used. Generator v2 differs from v1 only in how text windows are sampled (first the opening word type, then the rest); it was written to fix a missed text-beginner target and did not fix it. Both generators meet the size and length targets in every corpus and the remaining targets at similar, not identical, rates (all targets met by 13% of generator-v1 and 3-7% of generator-v2 language corpora at the Indus point, sister-v2 runs; full table in the appendix).

3.9 Sensitivity sweep

Duplicate-text rate and sign inventory are set exactly and varied independently on a 6 by 5 grid (duplicates 0-0.5, inventory 400-800) at the Indus point, with methods frozen. Duplicates are set by assembling a corpus from a larger pool of distinct texts plus copies weighted by popularity; inventory is set through the allograph rate alone, by bisection. Because inventory is raised only through allographs and no tested method merges allographs, the inventory axis measures robustness to unmerged variants, not to a larger underlying sign system; every map states this. A cell counts only if its corpora hit both settings within tolerance without distorting text lengths. The plausible box (duplicates 0.2-0.4, inventory 400-700) brackets the published duplicate rates and sign-list sizes. It was written into the run profile after the sweep had started, after a test rendering of the first 24 records and a count of which cells were reachable, but before recovery was tabulated for the sweep; it was first committed together with the results (Appendix: analysis history). Published duplicate rates and sign-list sizes are marked on the map as reference points, not targets.

4. Results

Labels used in this section. **Primary**: original (frozen) rule, scored before the cognate step, sister-v2. **With oracle sound correspondences**: the same runs scored after the cognate step. **Upper bound (sister-v1, known overlap)**: runs made before the sister split was corrected; these were scored only after the cognate step. The sensitivity sweep was run under both; the sister-v2 sweep gives the primary numbers in Sections 4.4 and 4.5.

4.1 Entropy statistics fail at Indus scale

This result does not depend on the sister language: Task A uses no reference language.

At the Indus point, the plug-in ratio of conditional to unigram entropy over the full sign alphabet is 0.470 for an i.i.d. 420-sign control and 0.477 for the synthetic languages (10th-90th percentile 0.35-0.67); fresh seeds give 0.467 and 0.476. With Rao's merge to the 100 most frequent signs, the i.i.d. control scores higher than the languages (0.696 vs 0.605). At this corpus size the statistic cannot tell language from independent signs.

The classifier built on it separates languages from controls with balanced accuracy 0.77 [0.57, 0.90] when the generator is calibrated to every target (full). When the frequency

and positional targets are left free (holdout), it falls to 0.57 [0.32, 0.81], and to 0.49 [0.25, 0.75] on fresh seeds; both intervals include chance. Under full, every Task A method classifies at least one structurally non-linguistic control family as language in all its corpora (for the entropy classifier, the rigid control of Rao et al. 2009). We therefore treat language detection at Indus scale as unresolved, and the full score as a product of calibration.

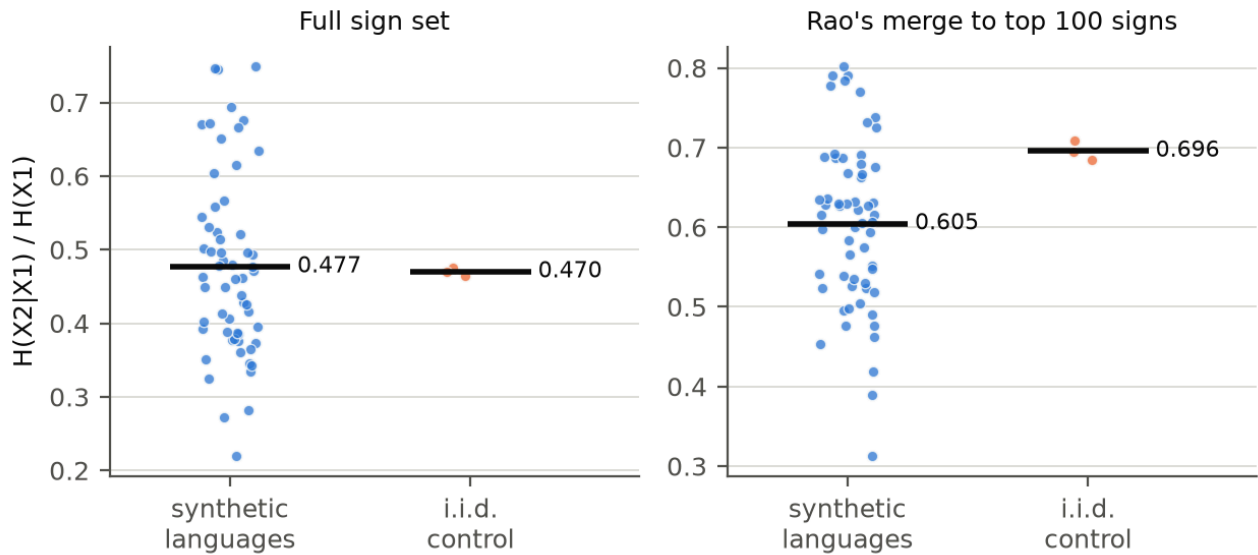


Figure 3. Entropy ratio $H(X2|X1)/H(X1)$ at the Indus point (2,906 texts, mean 4.6 signs; full calibration, seeds 0-2), left on the full sign set, right after Rao's merge to the 100 most frequent signs. Dots are corpora (60 synthetic-language corpora, 3 i.i.d.-control corpora); bars are means. Score label: not applicable; Task A uses no reference language, so no sister or cognate step is involved.

4.2 Without a related language

This result does not depend on the sister language: the hidden language and its sister are both excluded from this tier, so no cognate step runs.

Without a related language among candidates, mean recovery stays at or below 4.8% in every run; no corpus reaches 50% of tokens; the best single corpus reaches 28%.

Run	Mean, original rule	Mean, revised rule (post hoc)	Best single corpus (original / revised)	Corpora $\geq$ 50%
Generator v1, seeds 0-2	1.8%	3.3%	16.4% / 18.1%	0 of 60
Generator v1, seeds 3-5	2.4%	3.0%	19.7% / 23.8%	0 of 60
Generator v2, seeds 0-2	3.3%	4.8%	23.2% / 27.6%	0 of 60
Sensitivity sweep (sister-v1 hidden split), all valid cells	at most 4.6% per cell	at most 7.1% per cell	24.6% / 27.9%	0 of 1,571
Sensitivity sweep (sister-v2 hidden split), all valid cells	at most 4.6% per cell	at most 6.3% per cell	26.5% (either rule)	0 of 1,451

Inside the plausible box the no-relative mean is at most 4.3% (fixed-composition panel: the same 10 language-script combinations in every box cell), never above 7.1% anywhere on the grid (sister-v2 sweep: 4.3% and 6.3%). Longer texts do not change this: at 20 signs per text the no-relative tier stays at or below 6.5%.

4.3 Longer inscriptions vs more inscriptions

Under the frozen rule, longer inscriptions help more than more inscriptions in both generators. At equal total tokens (2,906 texts of 10 signs vs 6,317 texts of 4.6 signs, about 29,000 tokens each, both run directly), candidates tier:

Generator	Score	Longer	More	Difference, paired [95% CI]
v1	Primary	18.0%	2.6%	15.4 [5.8, 24.9]
v2	Primary	35.6%	18.3%	17.4 [8.5, 25.0]
v1	With oracle sound correspondences	28.6%	2.7%	25.9 [10.5, 41.4]
v2	With oracle sound correspondences	54.9%	27.6%	27.4 [12.8, 39.0]

On fresh seeds (size interpolated on log tokens, no interval): v1 17.7% vs 3.4%, v2 32.2% vs 11.8% (primary). The advantage holds from 10 signs per text in generator v1 and from 6 in generator v2. Without a relative, both stay at or below 6.5%, with a small edge for length. The result also holds for logo-syllabic corpora alone: 12.7% vs 0.2% (v1) and 27.8% vs 6.4% (v2), primary (Section 4.7).

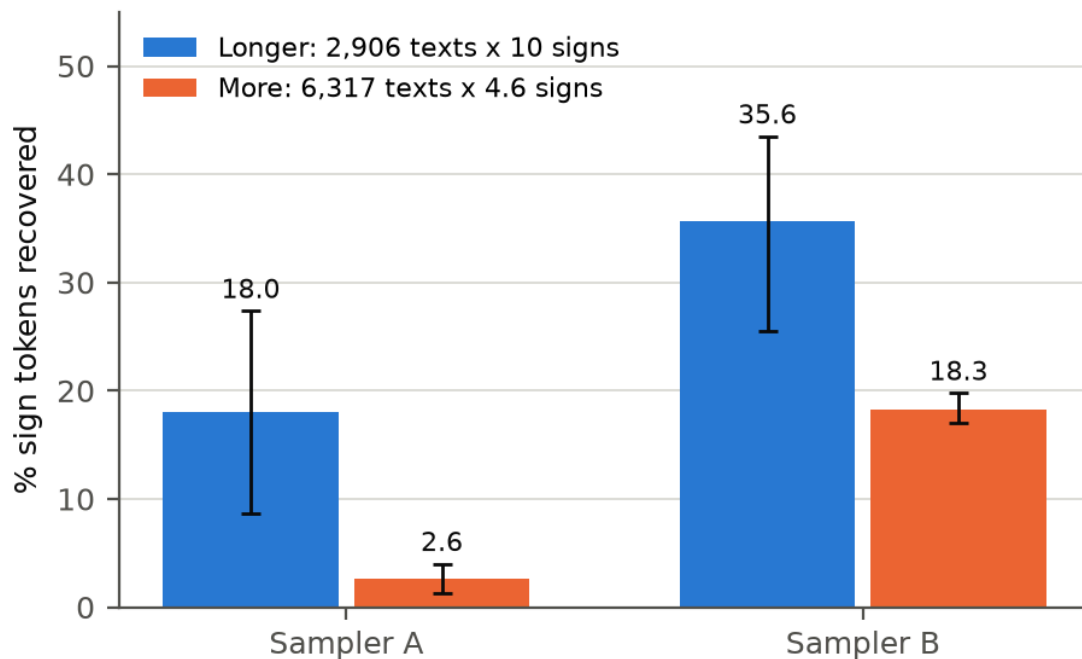


Figure 2. Longer vs more inscriptions at equal total tokens (about 29,000), candidates tier, original rule, both generators (sampler A = generator v1, sampler B = generator v2). Score label: primary (sister-v2, before the cognate step). Bars are means over 60 corpora; whiskers are 95% intervals resampling whole source languages.

4.4 Difficulty is not fixed by published Indus statistics

Published Indus statistics do not fix how decipherable an Indus-scale corpus is. Two generators that both match them differ roughly three- to fivefold when a related language is available. For logo-syllabic corpora, the most Indus-relevant type, this contrast appears only when the related language is the sole reference; among candidate languages it is unresolved.

Candidates tier, Indus point	Generator v1	Generator v2	Ratio
Primary, seeds 0-2	2.4% [1.2, 3.8]	12.9% [7.8, 17.4]	5.3x
Primary, seeds 3-5	2.7% [1.2, 3.8]	9.4% [4.7, 14.0]	3.5x
Primary, sister-v3 (Section 4.6)	2.6% [1.2, 4.2]	9.3% [5.4, 13.5]	3.7x
With oracle sound correspondences, seeds 0-2	2.8%	18.9%	
Upper bound (sister-v1, known overlap), seeds 0-2	2.1%	15.0%	

No generator-v1 corpus reaches 50% of tokens in either seed set (primary); 6 of 120 generator-v2 corpora do. For logo-syllabic corpora the candidates-tier contrast is unresolved (0.6% vs 3.6%, and 0.9% vs 1.6% on fresh seeds, with overlapping intervals), because the original rule seldom selects the sister for them; with the sister as the sole reference the contrast is clear (3.6% vs 18.8%, and 3.2% vs 20.4%; Section 4.7). In the related tier the contrast is smaller in ratio but larger in points (primary, seeds 0-2: 15.9% vs 30.2%).

The two generators differ only in how text windows are sampled. Both meet the size and length targets in every corpus, and neither meets every target in more than 13% of language corpora (sister-v2 runs), so the published statistics do not choose between them. Corpus size also behaves differently: from 500 to 50,000 texts, generator v1 rises only from 2.5% to 3.7% (primary), while generator v2 rises from 3.5% to 18.7%.

Within one generator the duplicate rate and inventory matter less under the frozen rule. Across the plausible box (duplicates 0.2-0.4, inventory 400-700), generator-v1 recovery stays at 2.9-4.2% (primary; box panel: the 10 language-script combinations valid in every box cell), and the median within-combination spread is 2.1 points (maximum 7.5). Upper bound (sister-v1, known overlap), with oracle sound correspondences: 3.5-5.1%, median spread 2.1, maximum 7.6. Generator-v2 corpora whose measured duplicate rate and inventory fall inside the same box (37 corpora, seeds 0-5) average 11.9% (primary; best 72.0%), so the box does not bound recovery across generators.

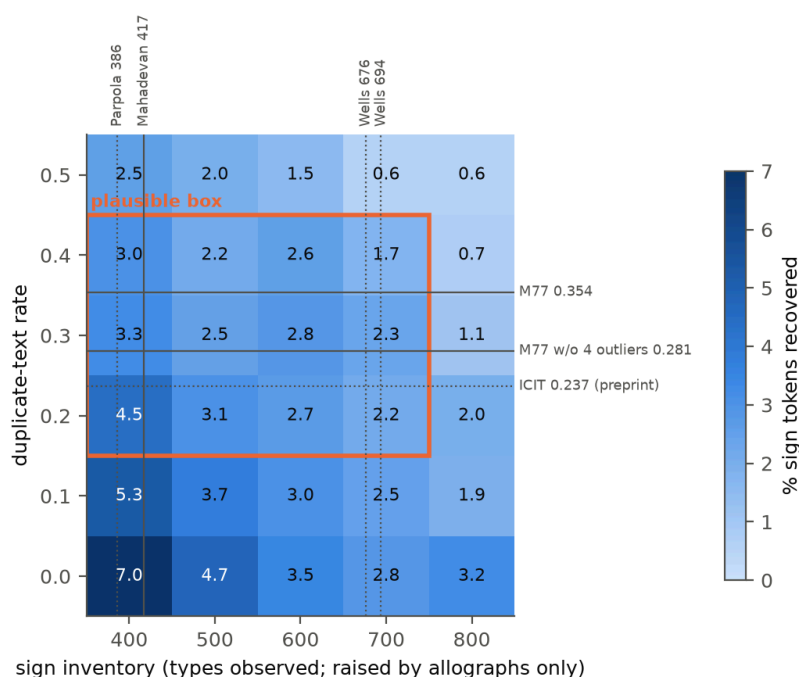


Figure 1. Sensitivity map, generator v1 (sampler A), candidates tier, original rule: mean % of sign tokens recovered over all valid corpora in each cell. Score label: primary (sister-v2, before the cognate step; no oracle sound correspondences). Orange: the plausible box. Lines: published sign-list sizes (top) and duplicate rates (right); solid lines are verified, dotted lines are read via a secondary source or come from an unverified preprint. Inventory is raised by allographs only, and no tested method merges them. The 2.9-4.2% range in the text is the box panel (the 10 combinations valid in every box cell); this figure shows all valid corpora. Upper-bound (sister-v1) maps: reports/sensitivity/appendix_sister_v1/.

Secondary, post hoc (revised rule): primary scores 9.1% (v1) vs 25.1% (v2); within the v1 box the revised rule ranges 5.6-14.2% (primary; upper bound 8.7-21.8%) and moves by up to 44 points within one language and script (upper bound 62).

4.5 Archaeological predictions (prediction, not measurement)

No seals-only, tablets-only or single-period corpus was tested. The positions below place qualitative published statements on the sensitivity map: S, the seals-only region, is the column with no duplicates (from "almost all ... unique" seals; Kenoyer & Meadow 2010); T, the tablets-only region, is the columns with duplicates 0.4-0.5 (at or above the pooled M77 rate of 0.354, a derived lower bound). Both span inventories 400-800. Numbers are primary (sister-v2 sweep, original rule, before the cognate step, generator v1) unless labelled.

Recovery in S is 2.8-7.0% against 0.6-3.0% in T, and S is higher at every inventory. In the strict panel (the 7 language-script combinations valid in every cell) S is higher only at 400, 500 and 800 signs (4.3% vs 0.6% at 400) and within 0.4 points of T at 600 and 700, so under the frozen rule the seals-only advantage is concentrated at small inventories. Under the revised rule (post hoc) S is 6.5-24.4% against 2.5-11.7%, higher at every inventory. Upper bound (sister-v1, known overlap), with oracle sound correspondences: S 3.5-14.5% vs T 0.4-4.9%. Without a related language every region stays at or below 6.3% (upper bound 7.1%).

A single-period corpus (fewer signs than the pooled 400-450; Kenoyer 2020b, p. 249) falls below the smallest inventory we can build, so the map cannot place it. Every subset also holds fewer than 2,906 texts.

4.6 Robustness: a sister with no shared hidden-text sequences (sister-v3)

Sister-v3 starts from sister-v2 and, for each corpus, drops every sister clause that contains any hidden text's word sequence. Verbatim containment is then zero by construction. The filter removes 33% (v1) and 42% (v2) of sister clauses at the Indus point, and the share of hidden-text word pairs still present in the sister falls from 31-35% to 7-8% (Sumerian keeps the most, 19-22%). Seeds 0-2, both generators, Indus point and the equal-token comparison; 480 corpora.

Indus point	v1, sister-v2	v1, sister-v3	v2, sister-v2	v2, sister-v3
related, primary	15.9%	12.4%	30.2%	27.3%
candidates, primary	2.4%	2.6%	12.9%	9.3%
candidates, with oracle sound correspondences	2.8%	2.6%	18.9%	12.3%

Headline 4 (Section 4.4) holds: 2.6% [1.2, 4.2] vs 9.3% [5.4, 13.5], 3.7x, intervals not overlapping. Headline 3 (Section 4.3) holds: longer beats more by 13.6 [5.8, 21.3] points (v1) and 24.1 [11.9, 35.9] (v2), primary. The filter depends on the corpus: many short texts remove more of the sister (40-52% of clauses) than fewer long texts (9-15%), which favours the long-text condition. Sister-v3 therefore confirms that the length advantage survives; it does not measure its size better than sister-v2.

The original rule chose the sister about as often under sister-v3 as under sister-v2 (v1: 20% vs 20% of corpora; v2: 28% vs 33%), and related-tier scores fell by about 3 points. Most of what the solver recovers from the sister therefore does not come from shared exact phrases.

4.7 Logo-syllabic corpora (most Indus-relevant script type)

"It is generally agreed that the signs found on seals and pottery represent a logosyllabic (morphemic) system" (Kenoyer 2020b, p. 249). The headline numbers restricted to logo-syllabic corpora (5 languages x 3 seeds = 15 corpora per point; primary unless stated):

Headline	Generator v1	Generator v2	Holds for logo-syllabic?
4.1 Entropy ratio, Rao top-100 merge (i.i.d. control 0.696)	0.591 (fresh seeds 0.587)	-	Yes: the i.i.d. control scores higher than the languages
4.2 No relative: mean, original / revised rule	0.0% / 0.8% (fresh seeds 0.1% / 0.9%)	0.3% / 1.1%	Yes: best single corpus 4.1%; none reaches 50%
4.4 candidates, seeds 0-2	0.6% [0.0, 1.5]	3.6% [0.2, 9.3]	Not resolved: intervals overlap
4.4 candidates, seeds 3-5	0.9% [0.0, 2.6]	1.6% [0.2, 3.9]	Not resolved
4.4 candidates, sister-v3	1.1% [0.0, 3.2]	3.4% [0.2, 9.2]	Not resolved
4.4 related, seeds 0-2	3.6% [2.4, 4.7]	18.8% [6.6, 31.0]	Yes
4.4 related, seeds 3-5	3.2% [2.0, 4.4]	20.4% [8.3, 34.8]	Yes
4.3 Longer vs more at ~29k tokens, candidates	12.7% vs 0.2%	27.8% vs 6.4%	Yes

For logo-syllabic corpora, recovery in the candidates tier is near zero in both generators, because the original rule selects Sumerian for almost every logo-syllabic corpus: it chooses the sister for 3 of 15 corpora (v1) and 4 of 15 (v2), mostly when the hidden language is Sumerian itself. The generator contrast appears only when the sister is the sole reference (related). With oracle sound correspondences the related contrast grows (5.7% vs 35.6%). Full table: [reports/sister_v2/logosyllabic_headlines.md](#).

4.8 Genre check: Sumerian seal inscriptions only

Four of the five sources are literary. As a genre check, Sumerian corpora were built only from the 19,076 Ur III seal inscriptions in the CDLI dump, for both the hidden corpus and its sister (Indus point, seeds 0-2, four script types, 12 corpora per generator), and compared with Sumerian corpora built from the full source.

Primary unless stated	v1, seal only	v1, full source	v2, seal only	v2, full source
related	13.1%	5.0%	8.8%	13.7%
candidates	9.0%	5.0%	7.7%	13.7%
candidates, with oracle sound correspondences	26.1%	6.8%	22.8%	17.2%
none, original rule	1.5%	2.2%	3.6%	2.3%

Recovery from seal-only material is of the same order as from the full source, with no consistent direction across generators, and the no-relative tier stays low (means at most 3.6%; best single corpus 20.2%). The check is limited: seal texts are far more repetitive and use fewer sign types than the calibrated corpora (duplicate rate 0.51-0.59, about 340 sign types, 37 signs covering 80% of tokens, against targets of 400-700 types and 69), and no seal-only corpus meets every calibration target, because the knobs were calibrated on the full source. The oracle step also adds more on seal-only material (15-17 points) than on the full source. The comparison is consistent with the main conclusions not depending on literary genre for Sumerian, but it is not a calibrated Indus-point result and cannot settle the question. Full table: `reports/sister_v2/sumerian_seal_genre_check.md`.

4.9 Positive controls

EM on English letter-substitution ciphers. Knight et al. (2006, Sec. 3) decipher a 417-letter encyclopedia article enciphered by a 1:1 letter substitution, with an English letter-bigram model in which word boundaries are known. Their bigram EM/Viterbi baseline makes 68 errors (83.7% correct); their best configuration (trigrams, cubed channel probabilities, smoothing) reaches 97.6%. We ran the frozen EM at benchmark settings (3 restarts, 60 iterations) on the same kind of task: a plaintext model from *Pride and Prejudice* and *Moby Dick* (70,000 or 1.5 million characters, matching their two data sizes), ciphertext from *On the Origin of Species*, word boundaries given, 10 random passages and keys per length (Project Gutenberg texts).

Cipher length (letters)	100	200	417	1,000	2,000	5,000
Original rule, 1.5M-character model	58.2%	92.2%	97.5%	99.7%	99.9%	99.9%
Original rule, 70,000-character model	71.7%	84.6%	97.4%	99.8%	99.9%	99.9%
Revised rule, 1.5M-character model	60.9%	92.7%	97.7%	99.6%	99.7%	99.8%

At the published length of 417 letters the frozen EM decodes 97.4-97.7% of letters, above the published bigram baseline and level with the published best configuration. The settings are close but not identical: Knight et al. decode with Viterbi over the whole text, while our solver assigns one value per cipher letter, and our texts are novels and Darwin rather than news and an encyclopedia. The solver therefore works on the standard problem it was built for, and its low recovery on Indus-like corpora is not explained by a broken implementation.

Conditional-entropy ordering (Rao et al. 2009). Rao et al. report their values only as figures (Fig. 1A, conditional entropy against the number of most frequent tokens; Fig. 1B, conditional entropy relative to a random sequence), read here by eye; the supplement tabulates only Indus perplexities. Using the frozen entropy function on our own reference corpora at matching unit levels, with a shuffled sequence as the random reference:

Corpus	Rao et al., Fig. 1B (by eye)	Ours, relative, 400 tokens	Ours, nats, 400 tokens
Random-order control (their type 1)	1.00	1.00	5.45
Sanskrit	0.66	0.75	3.35
English words	0.64	0.81	3.31
Sumerian	0.57	0.67	3.24
Old Tamil	0.56	0.74	3.34
English characters	0.51	0.85	2.47
Rigid control (their type 2)	about 0	0.02	0.08

The extremes reproduce, and at 400 tokens our language corpora sit at 3.2-3.4 nats against roughly 3.3-3.8 in Rao's Fig. 1A. The ordering among languages does not reproduce: English characters rank highest for us and lowest for Rao, and Sanskrit is above Sumerian in both but by different margins. Our corpora (Ramayana rather than Rig Veda, CDLI rather than ETCSL, Pride and Prejudice rather than the Brown corpus), unit definitions and random reference differ, and English characters have fewer than 400 types, so this is a partial reproduction only. Files: reports/positive_controls/.

5. Limitations

1. **Allograph merging.** Sign inventory is raised only through allographs, and no tested method merges them. The inventory axis measures robustness to unmerged variants.
2. **Sister-language distance and closeness.** The sister keeps word order and morphology, comes from the same source text (verbatim overlap 30-56% by language under sister-v2), and is paired with oracle sound correspondences. Its distance is a free parameter. Sister tiers are optimistic.
3. **Oracle script type and unit level in Task D.** Solvers are told the script type; real decipherers are not.
4. **Positive controls are partial.** The frozen EM matches the best published accuracy on English letter-substitution ciphers (Section 4.9). The entropy statistic reproduces the extremes and the magnitude reported by Rao et al. (2009) but not their ordering among languages. No positive control exists for the EM cognate-matcher or for the n-gram, positional, segmentation and script-type methods, so a low score from those could still reflect our implementation.
5. **Genre.** Four of the five sources are literary (an epic, classical poetry, Caesar and Vergil, an oral epic), cut into short seal-like clauses. Only Sumerian includes real seal inscriptions. Indus texts are mostly short inscriptions on seals and tablets. A genre check on Sumerian seal-only corpora (Section 4.8) gives recovery of the same order as the full source, but those corpora miss the calibration targets.
6. **One language, one fixed script per corpus.** The Indus script was used for about 700 years and changed over that time (Kenoyer & Meadow 2010, p. 8), and it may have been used to write more than one language (Kenoyer 2020b, pp. 237, 240, 254). Every synthetic corpus encodes a single language in a single fixed script. A pooled corpus that mixes languages or stages of the script is not modelled, and could be harder than anything tested here.

7. **Few clusters.** Intervals resample five source languages. With so few clusters a percentile cluster bootstrap can produce intervals that are too narrow; we did not measure their coverage, so every interval should be read as approximate. Logo-syllabic subsets (Section 4.7) have only 15 corpora per point.
8. **Text-beginner calibration.** Most corpora miss the text-beginner target (32-37% meet it, sister-v2 runs), and generator v2 did not fix this.
9. **Median length is derived, not published.** The target of 4 is derived from Yadav et al. 2010, Fig. 2 (2,591 texts, mean 3.92, max 14), which is not like-for-like with 2,906 texts at 4.6 signs; it was originally unsourced. It is used, not only checked: it sets the spread of the generated length distribution, through $\sigma = (\ln 17 - \ln 4) / z$. Removing it from the calibration objective would change nothing, because every corpus already meets it and only unmet targets enter the objective. Removing it from the generator would require another assumption for the spread, whose effect is unknown without rerunning.
10. **CDLI licence.** Academic reuse with citation, not an open licence; Sumerian is excluded from published challenge rounds.
11. **The 2,591 vs 2,906 discrepancy** in the source of the M77 duplicate rate and the median.
12. **Subset positions are estimated** from qualitative statements and raw counts.
13. **The single-period subset cannot be placed;** every subset is smaller than 2,906 texts.
14. **Post-hoc decisions.** The revised selection rule, generator v2, the sister corrections and several analyses were added after results were seen. Every such decision is listed in the analysis-history appendix. The original rule is primary throughout.
15. **Plug-in entropy is biased at this scale,** so any fixed entropy cut-off depends on corpus size.
16. **Controls built by us.** A classifier is only as general as its controls.
17. **No neural decipherment model is evaluated.** The EM cognate-matcher contains no neural network.

6. Blind challenge and reproducibility

The challenge has public rounds, which release corpora and keys, and hidden rounds, which release corpora and SHA-256 commitments of their keys. Hidden keys are generated and held by the maintainer and are not in the repository. Challenge corpora are written from secretly disguised sister languages, so they cannot be matched against the public source texts. Each team gets one scored submission per hidden round. Sumerian-derived corpora are excluded from all published rounds. The static leaderboard marks entries whose authors claim a real Indus decipherment.

Every number in this paper can be regenerated from the repository. Method code is frozen at the tag `frozen-v1` and the data release at `data-v1`; later changes (sister-v2, sister-v3 and the analysis scripts) touch only data generation and analysis and are selected per run profile. Each run records its generator and sister version. Source texts are fetched by the user under their own licences and are never redistributed.

7. Conclusion

On synthetic corpora that match the published Indus statistics, mean recovery without a related language stays at or below 4.8% in every run, and longer inscriptions help more than more inscriptions. Published Indus statistics do not fix how decipherable an Indus-scale corpus is: two generators that both match them differ roughly three- to fivefold when a related language is available, and for logo-syllabic corpora, the most Indus-relevant type, this contrast appears only when the related language is the sole reference; among candidate languages it is unresolved. A statement about how hard the Indus script is to decipher therefore has to name its corpus (object types, period, duplication), its knowledge tier, its script type and its selection rule. None of these results bears on what the Indus signs mean.

Appendix A. Analysis history

Every decision taken after any result was seen, in order, with the commit that records it. Decisions taken before any result are marked "pre-results". Dates are 2026, UTC.

#	Date	Decision	Pre-results or post-results	Commit
1	3 Oct	Adversarial design review (Believer, Skeptic, Investor, Judge) fixes six safeguards: band and feasibility reporting, three calibration regimes, structural controls, knowledge tiers, "no method succeeds" regions reported, probe-resistant challenge	Pre-results	b1194b0
2	3 Oct	Calibration targets corrected from unverified project-brief values (5,500 texts, mean 4.4) to Mahadevan (1977) via Yadav et al. 2010 and Rao 2018 (2,906 texts, mean 4.60); all results regenerated. The median target (4) was kept from the brief, unsourced	Post-results (development runs)	05b91d3
3	3 Oct	Two EM corrections during development (pooled rare-unit state pinned; candidate selection by normalised bigram gain) produce the revised rule. The rule as first written is kept as the original rule and both are reported	Post-results (development runs)	e625f56
4	3 Oct	Method code frozen at frozen-v1; Sumerian excluded from published challenge rounds pending CDLI terms	Pre-final-run	e625f56
5	3 Oct	Final run, 1,554 corpora	-	b1f8704
6	3 Oct	Length sweep added to the replication profile; fresh-seed replication (seeds 3-5) run	Post-results	bbd0e49, d997015
7	3 Oct	Generator v2 written to fix the missed text-beginner target (did not fix it)	Post-results	7246062, fde7031
8	3 Oct	Sensitivity sweep over duplicate rate and inventory added; headline order changed to lead with generator dependence	Post-results	4dcb9a1
9	3 Oct	Plausible box (duplicates 0.2-0.4, inventory 400-700) written into the profile after a test rendering of the first 24 sweep records and a reachability count on 1,216 records, before sweep recovery was tabulated. First committed together with the sweep results, so the order rests on the session log, not on git	Partly pre-results (see text)	a0e1233
10	3 Oct	M77 duplicate rate (0.354) measured from the vector data of Yadav et al. 2010, Fig. 2, and marked on the map as a reference point	Post-results	a0e1233
11	4 Oct	After input from J. M. Kenoyer: "stray finds" and "secure context" wording removed; object-type and period notes added; archaeological subsets placed on the map as predictions	Post-results	00f243f, 30a7ec5
12	4 Oct	Original vs revised rule reanalysed (no new runs). The original rule made primary throughout; "original rule stays near 5% in the box" found to hold for generator v1 only	Post-results	8007ed7
13	4 Oct	Audit of the 93% corpus finds no key leak or scoring bug, but a sister clause-split defect and the size of the oracle cognate step	Post-results	900d104
14	4 Oct	Sister-v2 (split by clause content) and before-cognate scoring; sister-v1 numbers relabelled "upper bound (sister-v1, known overlap)"; primary redefined as original rule before the cognate step	Post-results	0a5c913, 87d5706
15	5 Oct	Sister-v3 robustness check and equal-token size points	Post-results	1cb6faf
16	5 Oct	Headline 3 wording changed from "roughly fivefold" to "roughly three- to fivefold" after replication (3.5x) and sister-v3 (3.7x)	Post-results	457be13
17	5 Oct	Residual-overlap figure corrected from 38-82% (one script type and seed, estimated under the old split) to 30-56% (12 corpora	Post-results	40acfba

#	Date	Decision	Pre-results or post-results	Commit
		per language, measured under sister-v2)		
18	5 Oct	Median target relabelled, first as "unsourced", then as "derived from Yadav et al. 2010, Fig. 2"; it was originally taken from the project brief without a source	Post-results	40acfba, this revision
19	5 Oct	Sensitivity sweep rerun under sister-v2 with before-cognate scores	Post-results	this revision
20	5 Oct	Logo-syllabic headline split and Sumerian seal-only genre check added	Post-results	this revision
21	5 Oct	One configuration typo caused 60 failed jobs in an equal-token profile; fixed and rerun, failed records discarded	Operational	1cb6faf
22	5 Oct	Headlines reordered (length vs size becomes headline 3; underdetermination becomes headline 4) and headline 4 reworded to state the logo-syllabic result	Post-results	this revision
23	5 Oct	Positive controls added: frozen EM on English letter-substitution ciphers, and conditional-entropy ordering against Rao et al. (2009) figures. The latter showed that our control keys number Rao's type 1 and type 2 the other way round	Post-results	this revision
24	5 Oct	Sensitivity sweep aggregated with before-cognate scores. Seals-only and tablets-only regions defined as duplicates 0.0 and 0.4-0.5 at all inventories, for both sweeps; an earlier tablets-only range (1.4-4.9%) had used the 0.4 column only	Post-results	this revision
25	5 Oct	Nair (2026) availability checked: on request only, no public repository or licence	-	this revision

Back matter

Data and code availability. Code, configurations, run records and reports are in the repository under the tags above; no third-party text is redistributed.

Acknowledgements. Code was developed with AI coding assistance.

Appendices. Analysis history (Appendix A); sweep maps and tables, primary (reports/sensitivity/) and upper bound (reports/sensitivity/appendix_sister_v1/); calibration, full result tables, sensitivity tables, the rule comparison, the audit, the sister-v1/v2/v3 comparisons and method deviations.

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